

BHARATHWAJ RAMANATHAN

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EDUCATION

University of Illinois Urbana-Champaign, USA

Aug 2024 – Dec 2025

Master of Engineering, Mechanical Engineering

GPA: 3.9/4.0

Related coursework: Additive Manufacturing and Product Design, Finite Element Analysis, Data-driven Design Methods

PSG College of Technology, Coimbatore, India

Jul 2020 - May 2024

Bachelor of Engineering, Mechanical Engineering

GPA: 8.59/10

Related coursework: Design of Machine Elements, Heat and Mass Transfer, Functional and Conceptual Design

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, Fusion360, ANSYS (Mechanical, Icepak), Hypermesh, Abaqus, nTopology

Programming and Tools: Python, MATLAB

Manufacturing & Analysis: FEA, Thermal and Structural Analysis, 3D Printing, Lattice Design, Prototyping, Filament Winding

WORK EXPERIENCE

VINJAMURI INNOVATIONS

New Rochelle, USA

Mechanical Engineering Intern

May 2025 – Present

- **Designing** and **analyzing** patented **Type III composite cylinders** rated for up to 750 bar using **SolidWorks** and **Abaqus**.
- **Simulating** thermal and structural performance under real-world loading, including pressure cycling from **160 to 750 bar**.
- **Researching hydrogen storage** technologies and transport standards for **sustainable energy infrastructure**.
- **Exploring** commercialization strategies for the **patented boil-off gas removal system** targeting long-duration storage.

INDUSTRY CAPSTONE – PRENAC TOOLS

Coimbatore, India

Project Intern

Dec 2023 – May 2024

- **Developed** and **optimized** a collapsible metal storage pallet using **SolidWorks**, reducing storage footprint by **40 percent**.
- **Conducted** structural simulations using **ANSYS Mechanical** to optimize the pallet's **1-ton load capacity**.
- **Enhanced** reusability and durability, reducing material replacement costs by **30 percent**.
- **Led** prototype fabrication and testing using **GD&T** and **ASME Y14.5** for dimensional accuracy and manufacturability.

DEFENSE RESEARCH AND DEVELOPMENT ORGANISATION

Bangalore, India

Project Trainee, Center for Airborne Systems

May 2023

- **Conducted finite element analysis** on an avionics mission system rack for airborne applications.
- **Used** shock mount simulations to maintain deformation under **3.13 mm** and stress under **238 MPa** at **2.56g**.
- **Validated** structural integrity under limit and emergency load conditions (up to **9g**).
- **Gained** experience in meshing, structural analysis, and aerospace material selection.

ACADEMIC PROJECTS

DESIGN AND SIMULATION OF COLD PLATE FOR SINGLE PHASE COOLING

University of Illinois Urbana Champaign – Research under faculty

Jan 2025 – May 2025

- **Designed** a compact cold plate with **BCC lattice-based heat sinks** using **SolidWorks** and **nTopology**.
- **Simulated** heat transfer and flow using **SolidWorks Flow Simulation** to compare multiple lattice configurations.
- **Evaluated** thermal and hydraulic performance to guide iterative design improvements.
- **Developed** design recommendations for future cold plate iterations based on multi-parameter tradeoff analysis.

RELIABILITY-BASED DESIGN OPTIMIZATION OF LATTICE COLD PLATE

University of Illinois Urbana Champaign

Jan 2025 – May 2025

- Built **Gaussian Process surrogate models** to predict max temperature and pressure drop in **gyroid lattices**.
- **Generated** design data using **Latin Hypercube Sampling** for varying lattice and flow configurations.
- **Applied Reliability-Based Design Optimization** to identify robust cold plate designs under uncertainty using **MATLAB** workflows.
- **Demonstrated** improved reliability under parameter uncertainty, optimizing for both thermal resistance and flow loss.

SINGLE PHASE IMMERSION COOLING SOLUTION FOR EV BATTERIES

University of Illinois Urbana Champaign

Aug 2024 – Dec 2024

- **Designed** and **simulated** an immersion-cooled battery system in **SolidWorks** and **ANSYS Icepak**.
- **Reduced** max temperature by **46.23 percent** and thermal variation from **23.7°C to 7.02°C**.
- **Identified** optimal coolant and reduced system pressure drop to improve energy efficiency and reliability.
- **Enhanced** safety margins by reducing hotspot severity and improving uniformity of temperature distribution.